University of Central Punjab, Lahore
Artificial Intelligence

BSCS

Assignment-02

CLO	CLO STATEMENT	Bloom's Taxonomy Level	PLO
2	To design and implement AI applications that solve real-world problems.	C3	P3

Learning Objectives

- **Apply and trace search algorithms** (BFS, DFS, UCS, Greedy, A*) to solve problems and understand their execution process.
- **Analyze and compare algorithm performance** in terms of completeness, optimality, and efficiency.
- **Evaluate heuristic functions and problem-solving approaches** for selecting appropriate search strategies in different scenarios.

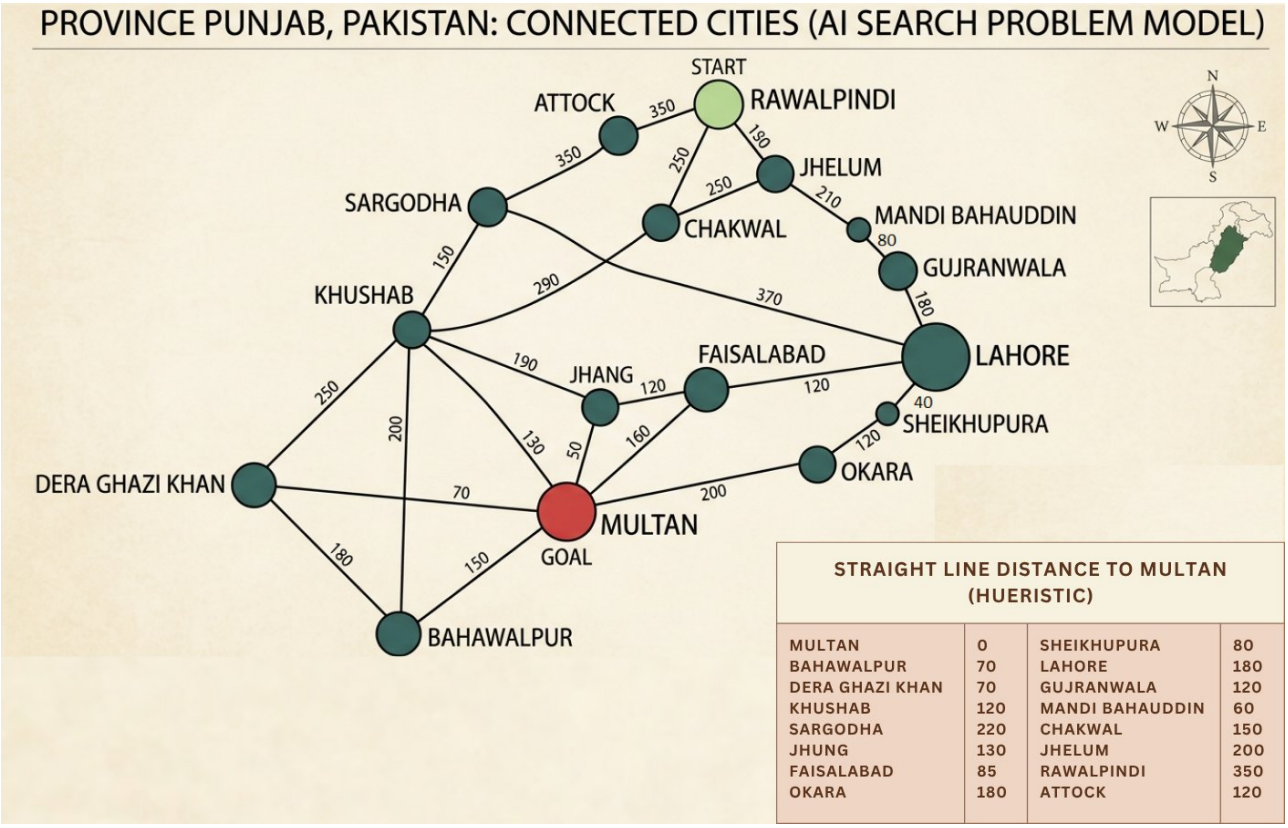
Instructions:

1. **Apply each search algorithm step-by-step** as demonstrated in class, clearly showing the evolution of the **OPEN and CLOSED lists**, node expansions, and path selection.
2. **Provide proper reasoning** for all analytical and conceptual questions. Answers without justification will not receive full marks.
3. **Assignment Type**

   This is an **individual assignment**. Each student must submit their own work.

4. **Plagiarism Policy**
   Plagiarism is strictly prohibited. Any submission found to be copied (from peers or external sources) will be awarded **zero marks**.
5. **Marks Distribution & Deadline**

Deliverables	Due Date	Parts	Marks	Sub-Total	Total
Deliverable	**Submission by 09:00 AM 10/11/2026.**	Part A	15	25	60
			10		
		Part B	10	10	
		Part C	10	10	
		Part D	5	5	
		Part E	10	10	



Part A: A* and Heuristics (25 Marks)

Q1. A* Execution (15 Marks)

Using above provided graph

Tasks:

1. Apply **A*** Search
2. Show:
   - 2.1. f(n) = g(n) + h(n)
   - 2.2. OPEN list updates
   - 2.3. Final path

Q2. Heuristic Analysis (10 Marks)

a) Is the given heuristic **admissible**? Justify
b) Is it **consistent**? Justify
c) What happens if heuristic overestimates?

Part B: State-Space Problem (10 Marks)

Q3. Water Jug Problem

You have:

- Jug A = 4 litres
- Jug B = 3 litres

Goal: Measure exactly **2 litres**

Tasks:

1. Define:
    - State representation
    - Operators
2. Solve using BFS (show steps)

Part C: Critical Thinking (10 Marks)

Q4. Algorithm Behaviour

Given a graph where:

- One path is very deep but cheap
- One path is shallow but expensive

Answer the following:

- Which algorithm prefers which path:
    - DFS
    - UCS
    - Greedy
    - A*

Explain briefly.

Part D:(5 Marks)

Q5. Debugging A*

A student computed:

f(n) = h(n) only

instead of:

f(n) = g(n) + h(n)

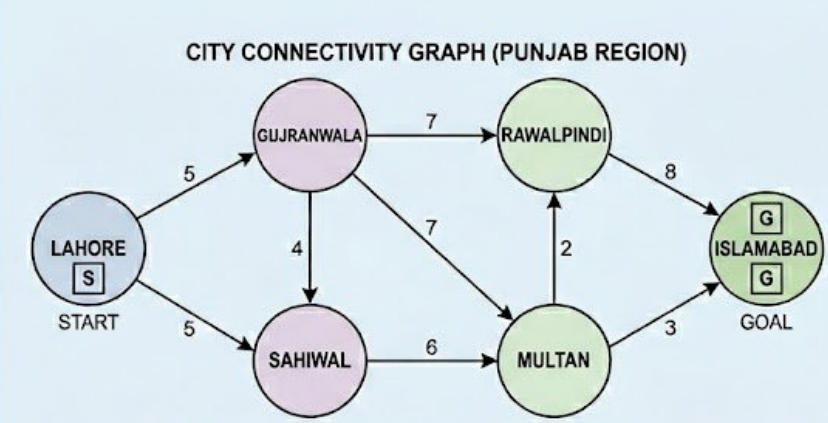
Task:

- Explain the mistake
- What algorithm does it become?

Part E: (10 Marks)

Trace the execution of the A* (A-Star) algorithm using the provided h(n) heuristic from LAHORE to ISLAMABAD.

(a) Show the state of the Frontier priority queue at each step, listing the f(n) values for each expanded node.



HEURISTIC FUNCTION h(n)

CITY (n)	h(n) [STRAIGHT-LINE DISTANCE TO ISLAMABAD]
LAHORE	12
GUJRANWALA	8
SAHIWAL	5
RAWALPINDI	4
MULTAN	3
ISLAMABAD	0